Shariq Mohammed

CURRICULUM VITAE

Department of Biostatistics
School of Public Health
Boston University
801 Massachusetts Avenue (CT303)
Boston, MA 02118

Email: shariqm@bu.edu
Phone: (617) 358-2682
Web: shariq-mohammed.github.io
Scholar: Google Scholar profile
ORCID: 0000-0002-3107-2969

ACADEMIC APPOINTMENTS

Boston University (BU), Boston, MA

Assistant Professor *Aug 2021–present*
Department of Biostatistics, School of Public Health (SPH)

Core Faculty *Oct 2026–present*
Center for Brain Recovery

Faculty Affiliate *Aug 2025–present*
Rafik B. Hariri Institute for Computing and Computational Science & Engineering

Rafik B. Hariri Junior Faculty Fellow *Aug 2022–Jul 2025*
Rafik B. Hariri Institute for Computing and Computational Science & Engineering

University of Michigan (U-M), Ann Arbor, MI

Postdoctoral Research Fellow *Sep 2018–Jul 2021*
Mentors: Veerabhadran Baladandayuthapani & Arvind Rao

Precision Health Scholar *Sep 2019–Jul 2021*

EDUCATION

University of Connecticut (UConn), Storrs, CT

Ph.D. in Statistics *Aug 2018*
Dissertation: *Bayesian variable selection with applications to neuroimaging data*
Advisors: Dipak K. Dey & Yuping Zhang

M.S. in Statistics *Sep 2017*

Chennai Mathematical Institute (CMI), Chennai, India

M.Sc. in Applications of Mathematics *May 2014*

Indian Statistical Institute (ISI), Bangalore, India

B.Math. (Hons.) *Jun 2012*

PRIOR RESEARCH EXPERIENCE

The Travelers Companies, Hartford, CT
Research Assistant *Aug 2016–Aug 2018*

University of Connecticut, Storrs, CT
Graduate/Research Assistant *Aug 2016–Aug 2017*

Pfizer Inc., Boston, MA
Summer Student Worker *Summer 2016*

Tata Consultancy Services, Innovation Labs, Hyderabad, India
Research Intern *Summer 2013*

RESEARCH INTERESTS

Methodology: Bayesian hierarchical modeling and variable selection; spatial and spatio-temporal statistics; functional and geometric data analysis; integrative analysis of multi-modal data; network-based modeling of complex data

Applications: Alzheimer's disease and related dementias (ADRD); digital cognitive assessments; neuropsychology; spatial omics; biomedical imaging and radiogenomics; population health and community well-being

RESEARCH FUNDING

Current Funding

Down Syndrome Toward Optimal Trajectories and Health using Medicaid Analytic eXtract (DS-TO-THE-MAX)
NIH/NIA, R01AG073179 *Sep 2026–Aug 2031*
Role: Co-Investigator (PI: Eric Rubenstein)
Total costs: $4,109,950

Scalable software engineering and methods for large-scale subcellular spatial omics
NIH/NHGRI, R01HG014487 *Sep 2025–Aug 2029*
Role: Co-Investigator / BU Site PI (PI: Ruben Dries)
Total costs: $3,371,645; BU subaward: $523,243
Scored in the 1st percentile on initial submission

Network biomarkers for early detection of cognitive impairment using digital clock drawings
NIH/NIA, R03AG098551 *Jan 2026–Dec 2027*
Role: Principal Investigator
Total costs: $323,825
Scored in the 2nd percentile on initial submission

Removing batch effects in high-throughput biomedical studies
NIH/NIGMS, R01GM127430 *Sep 2023–Aug 2027*
Role: Co-Investigator (PI: William E. Johnson)
Total costs: $1,151,799

Pending Funding

Interpretable statistical machine learning for digital trail making test-derived biomarkers of cognitive impairment in ADRD
NIH, R21 *Jul 2027–Jun 2029*
Role: Multiple Principal Investigator (MPI: Abolfazl Shafikhani)
Requested total costs: $458,708

EDEMA-ALERT: Evaluation of deterioration and edema monitoring using dynamic models in large ischemic stroke with Life Threatening Mass Effect
NIH, R01 *Jul 2027–Jun 2032*
Role: Co-Investigator (PI: Charlene J. Ong)
Requested total costs: **[amount]**

Tumor ecosystems and breast cancer outcomes in medically complex patients
NIH, R01 *Apr 2027–Mar 2032*
Role: Co-Investigator (MPIs: Ruben Dries and Mollie Bernard)
Requested total costs: $4,803,008

Framingham Study Ultra-Low-Field MRI: ADRD Imaging & Risk
NIH, R01 *Apr 2027–Mar 2032*
Role: Co-Investigator (PI: Hugo J. Aparicio)
Requested total costs: $5,302,815

***[title]** (Black Women's Health Study)*
NIH, U01 ***[period]***
Role: Co-Investigator, years 2–5 (MPIs: Kimberly A. Bertrand and Yvette C. Cozier)
Requested total costs: **[amount]**

Completed Funding

Longitudinal Risk Factor Changes and Early Recognition of Cerebral Small Vessel Disease
NIH/NINDS, R21NS135268 *Sep 2023–Aug 2026*
Role: Co-Investigator (PI: Jose R. Romero)
Total costs: $453,750

Mapping well-being and the social determinants of health
Sharecare, Inc., BU–Sharecare Contract *Nov 2021–Jul 2025*
Role: Co-Investigator (PI: Kimberly A. Dukes)
Total costs: $5,938,611

Predictive Utility of Pupillometry in Life-threatening Mass Effect: Management and Outcome Evaluation
BU Population Health Data Science Seed Award *Feb 2025–Jan 2026*
Role: Co-Investigator (PI: Charlene J. Ong)
Total costs: $5,000

Novel methods to detect spatially varying genes in spatial transcriptomics datasets
BU Population Health Data Science Seed Award *Feb 2024–Jan 2025*
Role: Principal Investigator
Total costs: $5,000

Developing novel digital neuropsychological test-based markers to quantify heterogeneity in participants' performance
Framingham Heart Study Brain Aging Program Pilot Award *Apr 2022–Mar 2023*
Role: Principal Investigator
Total costs: $25,000

Statistical modeling for topographic analysis of spatially resolved transcriptomics data
BU SPH Early Career Catalyst Award *Apr 2022–Aug 2023*
Role: Principal Investigator
Total costs: $19,782

Integrative decision models combining radiological-imaging and genotypic data in gliomas
U-M Precision Health Scholars Award *Sep 2019–Jul 2021*
Role: Principal Investigator
Total costs: $80,000

HONORS AND AWARDS

- *Rafik B. Hariri Junior Faculty Fellowship* ($20,000), Rafik B. Hariri Institute for Computing and Computational Science & Engineering, BU *2022–2025*
- *Precision Health Scholars Award* ($80,000), U-M *2019–2021*
- *Doctoral Dissertation Fellowship*, Graduate School, UConn *2018*
- *Doctoral Student Travel Award*, Graduate School, UConn *2017*
- *Multiple conference travel grants*, Department of Statistics, UConn *2017*
- *Pre-doctoral Dissertation Fellowship*, Department of Statistics, UConn *2016*
- *Matthew M. Goldstein Graduate Fellowship*, UConn *2015*

- *CMI Medal of Excellence* for outstanding performance in the M.Sc. in Applications of Mathematics program, CMI *2014*
- *Post-graduate Fellowship*, CMI *2012–2014*
- *INSPIRE Scholarship for Higher Education*, Ministry of Science & Technology, Government of India *2009–2014*
- *Undergraduate Fellowship*, ISI *2009–2012*

PUBLICATIONS

Key: Trainee = trainee author † = equal contribution * = (co-)corresponding author

Journal Articles

31. Jain, A., LaValley, M., Dukes, K.A., and **Mohammed, S.*** (2026). Modeling bounded well-being indices using Bayesian double generalized beta regression with spatial and temporal borrowing. *Spatial Statistics*, 76, 101047. doi:10.1016/j.spasta.2026.101047
Award: Eastern North American Region (ENAR) Distinguished Student Paper Award (2026)

30. Alperen, J.K., LaValley, M., Ni, P., Tripp, T., Emig, D., Patil, P., **Mohammed, S.**, Jain, A., Rickles, M., Colyer, E., Davis, B., Sullivan, L.M., Acker, M., and Dukes, K.A. (2026). Characterizing U.S. community well-being indices: a methodological synthesis and measurement systems perspective. *Population Health Metrics*. doi:10.1186/s12963-026-00502-4

29. Frady, G., Dey, D.K., and **Mohammed, S.*** (2026). Bayesian feature extraction using Gaussian and diffused-gamma priors for high-dimensional spatio-temporal data. *Statistics and Applications*, 24(2) (in press). Special issue on Recent Advances in Bayesian Statistics and Machine Learning. Journal article PDF

28. Shapiro, A.N., **Mohammed, S.**, Horsburgh, C.R., Jenkins, H.E., and White, L.F. (2026). Adapting back-calculation methods to estimate the incidence of tuberculosis. *Epidemiology*, 37(2), 220–227. doi:10.1097/EDE.0000000000001936

27. Dukes, K., Ni, P., Alperen, J., Cesare, N., LaValley, M., Tripp, T., Lane, K., **Mohammed, S.**, Patil, P., Emig, D., Winter, M., Davis, B., Wang, B., Jain, A., Acker, M., and Rickles, M. (2025). A national growth mixture modeling analysis of county-level COVID-19 incidence rate trajectories and health inequities during three successive pandemic waves in 2020. *Scientific Reports*, 15(1), 41272. doi:10.1038/s41598-025-25074-5

26. Song, J.J., Stafford, R.A., Pohlmann, J.E., Kim, I.S.Y., Cheekati, M., Dennison, S., Brush, B., Chatzidakis, S., Huang, Q., Smirnakis, S.M., Gilmore, E.J., **Mohammed, S.**, Abdalkader, M., Benjamin, E.J., Dupuis, J., Greer, D.M., and Ong, C.J.

(2025). Later midline shift is associated with better post-hospitalization discharge status after large middle cerebral artery stroke. *Scientific Reports*, 15(1), 11738. doi:10.1038/s41598-025-95954-3

25. Veerapaneni, D., Sakthiyendran, N.A., Du, Y., Mallinger, L.A., Reinert, A., Kim, S.Y., Nguyen, C., Daneshmand, A., Abdalkader, M., **Mohammed, S.**, Dupuis, J., Sheth, K., Gilmore, E.J., Greer, D., and Ong, C.J. (2025). Early pupil abnormality frequency predicts poor outcomes and enhances international mission for prognosis and analysis of clinical trials in traumatic brain injury (IMPACT) model prognostication in traumatic brain injury. *Critical Care Explorations*, 7(5), e1257. doi:10.1097/CCE.0000000000001257

24. Du, Y., Pohlmann, J., Chatzidakis, S., Brush, B., Mallinger, L., Stafford, R., Cervantes-Arslanian, A., Benjamin, E., Gilmore, E., Dupuis, J., Greer, D., Smirnakis, S., **Mohammed, S.**, and Ong, C. (2025). Quantitative Pupillometry predicts neurologic deterioration in patients with large middle cerebral artery stroke. *Annals of Neurology*, 97(5), 930–941. doi:10.1002/ana.27178

23. Gibbs, B., Sniderman, J., **Mohammed, S.**, Kain, M., Freccero, D., and Abdeen, A., on behalf of the PEPPER Investigators (2025). Association Between Tourniquet Use and Patient-Reported Outcomes Following Total Knee Arthroplasty: A Multicenter Comparison. *The Journal of Bone and Joint Surgery*, 107(9), 976–984. doi:10.2106/JBJS.24.00266

22. Jain, A., LaValley, M., Dukes, K., Lane, K., Winter, M., Spangler, K.R., Cesare, N., Wang, B., Rickles, M., and **Mohammed, S.**[*] (2024). Modeling health and well-being measures using ZIP Code spatial neighborhood patterns. *Scientific Reports*, 14(1), 9180. doi:10.1038/s41598-024-58157-w

21. **Mohammed, S.**, Masotti, M., Osher, N., Acharyya, S., and Baladandayuthapani, V. (2024). Statistical Analysis of Quantitative Cancer Imaging Data. *Statistics and Data Science in Imaging*, 1(1), 2405348. doi:10.1080/29979676.2024.2405348

20. **Mohammed, S.**[*], Kurtek, S., Bharath, K., Rao, A., and Baladandayuthapani, V. (2023). Tumor radiogenomics with Bayesian layered variable selection. *Medical Image Analysis*, 90, 102964. doi:10.1016/j.media.2023.102964

19. Romano, M.F., Zhou, X., Balachandra, A.R., Jadick, M.F., Qiu, S., Nijhawan, D.A., Joshi, P.S., **Mohammad, S.**, Lee, P.H., Smith, M.J., Paul, A.B., Mian, A.Z., Small, J.E., Chin, S.P., Au, R., and Kolachalama, V.B. (2023). Deep learning for risk-based stratification of cognitively impaired individuals. *iScience*, 26(9), 107522. doi:10.1016/j.isci.2023.107522

18. Halder, A.[†], **Mohammed, S.**[†], and Dey, D.K. (2023). Bayesian variable selection in double generalized linear Tweedie spatial process models. *New England Journal of Statistics and Data Science*, 1(2), 187–199. doi:10.51387/23-NEJSDS37

17. Chekouo, T., Stingo, F.C., **Mohammed, S.**, Rao, A., and Baladandayuthapani, V. (2023). A Bayesian group selection with compositional responses for analysis of radiologic tumor proportions and their genomic determinants. *Annals of Applied Statistics*, 17(4), 3013–3034. doi:10.1214/23-AOAS1749

16. Panigrahi, S., **Mohammed, S.**, Rao, A., and Baladandayuthapani, V. (2023). Integrative Bayesian models using post-selective inference: A case study in radiogenomics. *Biometrics*, 79(3), 1801–1813. doi:10.1111/biom.13740

15. Turk, S., Wang, N.C., Kitis, O., **Mohammed, S.**, Ma, T., Lobo, R., Kim, J., Camelo-Piragua, S., Johnson, T.D., Kim, M.M., Junck, L., Moritani, T., Srinivasan, A., Rao, A., and Bapuraj, J.R. (2022). Comparative study of radiologists vs machine learning in differentiating biopsy-proven pseudoprogression and true progression in diffuse gliomas. *Neuroscience Informatics*, 2(3), 100088. doi:10.1016/j.neuri.2022.100088

14. Qin, A., Lima, F., Bell, S., Kalemkerian, G.P., Schneider, B.J., Ramnath, N., Lew, M., Krishnan, S., **Mohammed, S.**, Rao, A., and Frankel, T.L. (2022). Cellular engagement and interaction in the tumor microenvironment predict non-response to PD-1/PD-L1 inhibitors in metastatic non-small cell lung cancer. *Scientific Reports*, 12(1), 9054. doi:10.1038/s41598-022-13236-8

13. Bhattacharyya, R., Banerjee, S., **Mohammed, S.**, and Baladandayuthapani, V. (2022). Spatial network-based modeling of COVID-19 dynamics: Early pandemic spread in India. *Journal of the Indian Statistical Association*, 60(1), 1–42. Journal article PDF

12. Krishnan, S.N.[†], **Mohammed, S.**[†], Frankel, T.L., and Rao, A. (2022). GaWRDenMap: A quantitative framework to study the local variation in cell-cell interactions in pancreatic disease subtypes. *Scientific Reports*, 12(1), 3708. doi:10.1038/s41598-022-06602-z

11. **Mohammed, S.**, Ravikumar, V., Warner, E., Patel, S.H., Bakas, S., Rao, A., and Jain, R. (2021). Quantifying T2-FLAIR mismatch using geographically weighted regression and predicting molecular status in lower-grade gliomas. *American Journal of Neuroradiology*, 43(1), 33–39. doi:10.3174/ajnr.A7341
Nominated: 2021 Lucien Levy Best Research Article Award, AJNR

10. Halder, A., **Mohammed, S.**, Chen, K., and Dey, D.K. (2021). Spatial Tweedie exponential dispersion models: An application to insurance rate-making. *Scandinavian Actuarial Journal*, 2021(10), 1017–1036. doi:10.1080/03461238.2021.1921017

9. **Mohammed, S.**[*], Bharath, K., Kurtek, S., Rao, A., and Baladandayuthapani, V. (2021). RADIOHEAD: Radiogenomic analysis incorporating tumor heterogeneity in imaging through densities. *Annals of Applied Statistics*, 15(4), 1808–1830. doi:10.1214/21-AOAS1458

8. **Mohammed, S.**[*] and Dey, D.K. (2021). Scalable spatio-temporal Bayesian analysis of high-dimensional electroencephalography data. *Canadian Journal of Statistics*, 49(1),

107–128. doi:10.1002/cjs.11592

7. Lee, J., Wang, N., Turk, S., **Mohammed, S.**, Lobo, R., Kim, J., Liao, E., Camelo-Piragua, S., Kim, M., Junck, L., Bapuraj, J., Srinivasan, A., and Rao, A. (2020). Discriminating pseudoprogression and true progression in diffuse infiltrating glioma using multi-parametric MRI data through deep learning. *Scientific Reports*, 10(1), 20331. doi:10.1038/s41598-020-77389-0

6. **Mohammed, S.***, Li, T., Chen, X.D., Warner, E., Shankar, A., Abalem, M.F., Jayasundera, T., Gardner, T.W., and Rao, A. (2020). Density-based classification in diabetic retinopathy through thickness of retinal layers from optical coherence tomography. *Scientific Reports*, 10(1), 15937. doi:10.1038/s41598-020-72813-x

5. Chekouo, T., **Mohammed, S.***, and Rao, A. (2020). A Bayesian 2D functional linear model for gray-level co-occurrence matrices in texture analysis of lower grade gliomas. *NeuroImage: Clinical*, 28, 102437. doi:10.1016/j.nicl.2020.102437

4. **Mohammed, S.***, Dey, D.K., and Zhang, Y. (2020). Classification of high-dimensional electroencephalography data with location selection using structured spike-and-slab prior. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, 13(5), 465–481. doi:10.1002/sam.11477
Invited: Statistical Analysis and Data Mining (SADM) Best Paper Session, Joint Statistical Meetings (JSM) 2022

3. Ray, D., Salvatore, M., Bhattacharyya, R., Wang, L., Du, J., **Mohammed, S.**, Purkayastha, S., Halder, A., Rix, A., Barker, D., Kleinsasser, M., Zhou, Y., Bose, D., Song, P., Banerjee, M., Baladandayuthapani, V., Ghosh, P., and Mukherjee, B. (2020). Predictions, role of interventions and effects of a historic national lockdown in India's response to the COVID-19 pandemic: Data science call to arms. *Harvard Data Science Review*, Special Issue 1. doi:10.1162/99608f92.60e08ed5

2. **Mohammed, S.***, Dey, D.K., and Zhang, Y. (2019). Bayesian variable selection using spike-and-slab priors with application to high dimensional electroencephalography data by local modelling. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 68(5), 1305–1326. doi:10.1111/rssc.12369

1. **Mohammed, S.*** and Dey, D.K. (2019). Assessing malaria using neutral-zone classifiers with mixture discriminant analysis on 2D images of red blood cells. *Journal of Biostatistics and Epidemiology*, 5(1), 1–11. doi:10.18502/jbe.v5i1.1901

Book Chapter

1. Matuk, J., **Mohammed, S.**, Kurtek, S., and Bharath, K. (2020). Biomedical applications of geometric functional data analysis. In *Handbook of Variational Methods for Nonlinear Geometric Data*, 675–701. Springer, Cham. doi:10.1007/978-3-030-31351-7_24

Conference Proceedings

3. Warner, E., Lee, J., Krishnan, S., Wang, N., **Mohammed, S.**, Srinivasan, A., Bapuraj, J., and Rao, A. (2023). Low-parameter supervised learning models can discriminate pseudo-progression and true progression in non-perfusion-based MRI. In *2023 45th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 1–4. doi:10.1109/EMBC40787.2023.10340435

2. Halder, A., **Mohammed, S.**, Chen, K., and Dey, D.K. (2022). Spatial risk estimation in Tweedie double generalized linear models. In *2022 Proceedings of International E-Conference on Mathematical and Statistical Sciences: A Selçuk Meeting*, 62–91. Proceedings PDF

1. Bhat, S.P., Kumar, M.U., and **Mohammed, S.** (2016). A dynamical systems approach to systemic risk in a financial network. In *2016 Indian Control Conference (ICC)*, 377–384. IEEE. doi:10.1109/INDIANCC.2016.7441162

Conference Abstracts

7. Manchanda, R., Shi, X., Pinheiro, A., Demissie, S., Aparicio, H., Lioutas, V., Ekenze, O., Beiser, A., DeCarli, C., Seshadri, S., Yang, Q., **Mohammed, S.**, and Romero, J.R. (2026). Lifetime Lipid Trajectories and Dementia in the Framingham Heart Study. *American Academy of Neurology 2026 Annual Meeting*. *Neurology*, 106(11 Suppl 1), S25.005. doi:10.1212/WNL.0000000000215344

6. Manchanda, R., Shi, X., Pinheiro, A., Demissie, S., Aparicio, H., Lioutas, V., Ekenze, O., Beiser, A., DeCarli, C., Seshadri, S., Yang, Q., **Mohammed, S.**, and Romero, J.R. (2026). Association Between Lifetime Lipid Trajectories and Stroke Prevalence in the Framingham Heart Study. *American Heart Association's International Stroke Conference 2026*. *Stroke*, 57(Suppl 1), DP060. doi:10.1161/str.57.suppl_1.DP060

5. Pinheiro, A., Demissie, S., Aparicio, H., Lioutas, V., Charidimou, A., Ekenze, O., Himali, J., Beiser, A., DeCarli, C., Seshadri, S., **Mohammed, S.**, and Romero, J.R. (2026). Life Stage-Specific Associations Between Vascular Risk Factor Trajectories and Cerebral Small Vessel Disease. *American Heart Association's International Stroke Conference 2026*. *Stroke*, 57(Suppl 1), DP003. doi:10.1161/str.57.suppl_1.DP003

4. Shi, X., Manchanda, R., Pinheiro, A., Demissie, S., Aparicio, H., Lioutas, V., Ekenze, O., Beiser, A., DeCarli, C., Seshadri, S., Yang, Q., **Mohammed, S.**, and Romero, J.R. (2026). Lifetime Lipid Trajectories in Relation to Post-Stroke Dementia Prevalence in the Framingham Heart Study. *American Heart Association's International Stroke Conference 2026*. *Stroke*, 57(Suppl 1), TP102. doi:10.1161/str.57.suppl_1.TP102

3. Pinheiro, A., Karjadi, C., Tripodis, Y., Kolachalama, V.B., Lunetta, K.L., Demissie, S., Liu, C., Au, R., and **Mohammed, S.** (2025). Predicting cognitive impairment using novel functional features of spatial proximity and circularity in the digital clock drawing test.

Alzheimer's & Dementia, 21(S6), e098476. doi:10.1002/alz70860_098476

2. Pinheiro, A., **Mohammed, S.**, Ekenze, O., Aparicio, H., Charidimou, A., Lioutas, V., Beiser, A., DeCarli, C., Seshadri, S., Demissie, S., and Romero, J.R. (2025). Lifelong trends in vascular risk factors and cerebral small vessel disease. *American Heart Association's International Stroke Conference 2025*. *Stroke*, 56(Suppl 1), WP285. doi:10.1161/str.56.suppl_1.WP285

1. Mallinger, L.A., Behm, C.F., Du, Y., Pohlmann, J.E., Gilmore, E.J., Greer, D.M., **Mohammed, S.**, and Ong, C.J. (2024). Quantitative Pupillometry Post-Surgical Decompression Predicts Severe Neurologic Disability in Patients with Acute Brain Injury. *22nd Annual Neurocritical Care Society Meeting*, San Diego, CA.

Submitted Manuscripts

7. Frady, G., Dey, D.K., and **Mohammed, S.**[*] (2026). Structured Gaussian and diffused-gamma feature extraction for electroencephalography data. Minor revision.

6. Pinheiro, A., Cheng, H., Karjadi, C., Gifford, K., Demissie, S., Lunetta, K., Liu, C., Au, R., and **Mohammed, S.**[*] (2026). Network representations of digital clock drawings for cognitive impairment detection. Minor revision.

5. Rampam, S., Perez Menendez, S.H., Jain, A., **Mohammed, S.**, Sepucha, K., and Abdeen, A.R. (2026). Impact of shared decision-making tools in patients' native languages on postoperative outcomes in total hip and knee arthroplasty at a safety-net hospital. Under review.

4. Spangler, K.R., Nori-Sarma, A., LaValley, M.P., **Mohammed, S.**, Wang, B., Cesare, N., Davis, B., Rickles, M., Dukes, K.A., and Lane, K.J. (2026). Euclidean distance, service-area networks, and population estimates: exposure measurement for grocery store access in U.S. metropolitan areas. Minor revision.

3. Wang, J., **Mohammed, S.**, Barocas, J.A., Bernson, D., White, L.F., and Kline, D.M. (2026). A spatial capture–recapture approach for estimating opioid use disorder prevalence in small areas using administrative data. Under review.

2. Pinheiro, A., Karjadi, C., Tripodis, Y., Kolachalama, V., Lunetta, K., Demissie, S., Liu, C., Au, R., and **Mohammed, S.**[*] (2026). Predicting cognitive impairment using novel functional features of spatial proximity and circularity in the digital clock drawing test. Under review. Preprint: doi:10.64898/2026.03.14.26348336
Award: ENAR Regional Advisory Board Poster Award to Adlin Pinheiro (2025)

1. Romero, J.R., Pinheiro, A., Demissie, S., Aparicio, H.J., Lioutas, V., Charidimou, A., Beiser, A.S., Ekenze, O., Himali, J.J., DeCarli, C., Seshadri, S., and **Mohammed, S.** (2026). Lifelong trajectories of modifiable vascular risk factors and relation to cerebral small vessel disease in the Framingham Heart Study. Under review. Preprint: doi:10.64898/2026.01.16.26344309

SOFTWARE

Open-Source R Packages on GitHub

- *BayesBadger*: Spatio-temporal beta regression (Journal Article 31) *2026*
- *clocknet*: Spatial networks from digital clock drawings (Submitted Manuscript 6) *2026*
- *marbles*: Multivariate Bayesian layered variable selection (Journal Article 20) *2022*
- *scalableBayesEEG*: Scalable spatio-temporal EEG analysis (Journal Article 8) *2020*
- *stSpikeSlabEEG*: Structured spike-and-slab EEG classification (Journal Article 4) *2020*
- *RADIOHEAD*: Tumor-heterogeneity radiogenomics (Journal Article 9) *2019*
- *SpikeSlabEEG*: Spike-and-slab variable selection for EEG data (Journal Article 2) *2019*

TEACHING

Instructor, Boston University

- *SPH BS 755*: Theory of Linear Models in Biostatistics *Fall 2025, Fall 2026*
- *SPH BS 704*: Introduction to Biostatistics *Fall 2025*
- *SPH BS 901*: Directed Studies in Biostatistics *Fall 2025*
- *CAS MA 575 / SPH BS 755*: Linear Models *Fall 2023, Fall 2024*
- *SPH BS 800*: Accelerated Statistical Training *Fall 2024*
- *SPH BS 730*: Introduction to `R`: Software for Statistical Computing *Spring 2022*

Instructor, University of Connecticut

- *STAT 3025Q*: Statistical Methods *Summer 2017, Fall 2017*

Short Courses and Workshops

- *Computational Biostatistics and Survival Analysis*, Tata Memorial Centre, Navi Mumbai, India (with Bhramar Mukherjee; course materials) *Dec 2019*
- *Data Visualization with R Shiny*, UConn *Apr 2018*

Teaching Assistant

- *Introduction to Statistics I & II*; *Introduction to Mathematical Statistics I & II*, Department of Statistics, UConn *Fall 2014–Spring 2016*
- *Numerical Linear Algebra*; *Probability Theory*, CMI *Spring 2013, Fall 2013*

STUDENT RESEARCH MENTORING

Doctoral Advisees

- Daniel Cho, PhD Biostatistics (Research Analyst; 2025–2026) *2026–present*
- Jordan Ahn, PhD Biostatistics *2025–present*

- Abhi Jain, PhD Biostatistics (dissertation co-advisor: Sara Lodi; research assistantship co-advisor: Kimberly Dukes) *2022–present*
  - *ENAR Distinguished Student Paper Award* *2026*
- Adlin Pinheiro, PhD Biostatistics *2022–present*
  - *BU SPH Pre-doctoral Research Ignition Award* *2022–2023*
  - *ENAR Regional Advisory Board (RAB) Poster Award* *2025*
  - *BU SPH Academic Travel Award* *2025*
  - *Alzheimer's Association International Conference Fellowship* *2025*
- Garrett Frady, PhD Statistics, UConn (co-advisor: Dipak K. Dey) *Graduated 2024*
  - Placement: *Mathematical Statistician, U.S. Food and Drug Administration*

Doctoral Dissertation Committees (PhD Biostatistics)

Current: Sophia Lu (2026–present), Jingxian Tang (2026–present)

Graduated: Evelyn Lauren (2026), Genevieve Dupuis (2026), Daniel Kojis (2026), Anne Shapiro (2026), Nimish Adhikari (2026), Jianing Wang (2023)

Master's Advisees

- Lin Tang, MS Biostatistics (Research Associate; joint with Huimin Cheng) *2026–present*
- Jude Sleiman, MPH *2024–2025*
  - Placement: *City Research Scientist II, Health Strategy Division, New York City Office of Labor Relations*
- Xiaohang Zhang, MS Biostatistics *2024–2025*
- John Cannon, MS Biostatistics *2022–2025*
- Yinzhu Piao, MA Statistics *2022–2024*
- Aaron Morris, MS Applied Biostatistics *Spring 2023*
- Kim-Judy You, MS Applied Biostatistics *Spring 2022*
- Tianxiang Sheng, MS Applied Biostatistics *Spring 2022*
  - Placement: *Biostatistician I, Department of Population Health Science and Policy, Mount Sinai*

MPH Integrated Learning Experience Advisees

- Jadon Demeritte, MacClement Guthrie, Romario Joseph, Deirdre Otoo, Neelsha Parekh *Fall 2026*
- Claudia Cheng, Anna Flynn, Lea Hur, Peijie Qiu *Fall 2025*
- Rishi Chigurupati, Cameron Pettinato, Morgan Powell, Katharine Reisdorf *Fall 2024*

- Alyssa Santoso, Olivia Seestadt *Fall 2023*
- Xinyi Du, Ian Hussain, Ziran Li, Shannon Mahowald, Alyson Speshock *Fall 2022*

TALKS

Invited Talks

49. Center for Brain Recovery Seminar, BU. *The shape of a clock: novel representations of digital clock drawing data for cognitive impairment prediction* *Sep 2026*
48. Joint Statistical Meetings, Boston, Massachusetts. *GASP: gradient-aware detection and localization of spatial patterns in gene expression* *Aug 2026*
47. Kyungpook National University, Statistics Seminar, Daegu, South Korea. *Modeling bounded well-being indices with Bayesian double generalized beta regression and spatial & temporal borrowing* *Jul 2026*
46. 33rd International Biometric Conference, Seoul, South Korea. *Modeling bounded well-being indices with Bayesian double generalized beta regression and spatial & temporal borrowing* *Jul 2026*
45. International Chinese Statistical Association China Conference, Shenzhen, China. *Modeling bounded well-being indices with Bayesian double generalized beta regression and spatial & temporal borrowing* *Jun 2026*
44. University of Massachusetts Amherst Statistics Seminar, Amherst, Massachusetts. *Borrowing structural information in Bayesian hierarchical models: applications to tumor imaging heterogeneity and geospatial well-being* *Apr 2026*
43. Framingham Heart Study–Brain Aging Program Seminar, BU. *Predicting cognitive impairment using novel functional features of spatial proximity and circularity in the digital clock drawing test* *Feb 2026*
42. ASA Biopharmaceutical Section Regulatory–Industry Statistics Workshop, Rockville, Maryland. *Digital neuropsychology, complex data, and the future of clinical trials* *Sep 2025*
41. BU Biostatistics Student Association Seminar, BU. *GitHub for R projects* *Apr 2025*
40. International Indian Statistical Association Annual Conference, Kochi, India. *Modeling well-being indices using ZIP code spatial neighborhoods under a Bayesian beta regression framework* *Dec 2024*
39. International Joint Conference on Computational and Financial Econometrics and Computational and Methodological Statistics (CFE–CMStatistics), London, United Kingdom. *Quantifying imaging heterogeneity via density functions with applications in brain and pancreatic cancer imaging* *Dec 2024*

38. Joint Statistical Meetings, Portland, Oregon (Topic-contributed). *GaWRDenMap: analyzing cell–cell interactions using spatial analysis and density functions to predict pancreatic disease subtypes* *Aug 2024*

37. Virginia Commonwealth University Biostatistics Seminar, Richmond, Virginia. *Quantifying imaging heterogeneity via density functions with applications in brain and pancreatic cancer imaging* *Apr 2024*

36. Connecticut College Research Seminar, New London, Connecticut (Virtual). *Quantifying imaging heterogeneity via density functions with applications in brain and pancreatic cancer imaging* *Apr 2024*

35. McGill University Biostatistics Seminar, Montreal, Canada (Virtual). *Quantifying imaging heterogeneity using density functions with applications in brain and pancreatic cancer imaging* *Mar 2024*

34. BU Biostatistics Student Association Seminar, BU. *GitHub for R projects* *Feb 2024*

33. Research on Tap on Health Data Science, Office of Research, BU. *Statistical methods for complex health data* *Nov 2023*

32. American Public Health Association Annual Meeting & Expo, Atlanta, Georgia (Roundtable session). *Modeling health and well-being measures by incorporating ZIP code spatial neighborhood patterns* *Nov 2023*

31. Federal University of Minas Gerais Mathematics and Statistics Seminar, Belo Horizonte, Brazil. *Layered variable selection for multivariate Bayesian regression with density function–based features: a case study in imaging genomics* *Aug 2023*

30. A New Era of Statistical Science: A Special Conference in Honor of Prof. Dipak Dey's 70th Anniversary, Belo Horizonte, Brazil. *GaWRDenMap: analyzing cell–cell interactions using spatial analysis and density functions to predict pancreatic disease subtypes* *Aug 2023*

29. Pushing the Boundary of Data Science through Statistical Modeling and Inference, Blacksburg, Virginia. *Layered variable selection for multivariate Bayesian regression with density function–based features: a case study in imaging genomics* *Jul 2023*

28. BU URBAN Research Methods Seminar, BU. *Modeling health and well-being measures by incorporating ZIP code spatial neighborhood patterns* *Apr 2023*

27. Framingham Heart Study–Brain Aging Program Seminar, BU. *Novel features from digital trail making and clock drawing tests for cognitive assessment* *Apr 2023*

26. BU Probability and Statistics Seminar, BU. *Layered variable selection for multivariate Bayesian regression using densities: a case study in imaging genomics* *Feb 2023*

25. International Indian Statistical Association Annual Conference, Bengaluru, India (Virtual). *Quantifying T2–FLAIR mismatch in gliomas using geographically weighted re-*

gression and density functions *Dec 2022*

24. BU SPH Think. Teach. Do. Spotlight Talk, BU. *Statistical methods for complex health data* *Nov 2022*

23. International Conference on Statistical Distributions and Applications, Huntington, West Virginia (Virtual). *Layered variable selection for multivariate Bayesian regression with density function–based features: a case study in imaging genomics* *Oct 2022*

22. BU Bioinformatics Seminar, BU. *Layered variable selection for multivariate Bayesian regression using densities: a case study in imaging genomics* *Oct 2022*

21. Royal Statistical Society International Conference 2022, Aberdeen, Scotland (Virtual). *Layered variable selection for multivariate Bayesian regression with density function–based features: a case study in imaging genomics* *Sep 2022*

20. Joint Statistical Meetings, Washington, DC. *Density-based classification in diabetic retinopathy using retinal layer thickness from optical coherence tomography* *Aug 2022*

19. Joint Statistical Meetings, Washington, DC (Invited poster; SADM Best Paper Session). *Classification of high-dimensional EEG data with location selection using a structured spike-and-slab prior* *Aug 2022*

18. 22nd Meeting of New Researchers in Statistics and Probability, George Mason University, Fairfax, Virginia. *Quantifying T2–FLAIR mismatch in gliomas using geographically weighted regression and density functions* *Aug 2022*

17. 35th New England Statistics Symposium, UConn. *Quantifying T2–FLAIR mismatch in gliomas using geographically weighted regression and density functions* *May 2022*

16. BU Epidemiology Seminar, BU. *Layered variable selection for multivariate Bayesian regression: a case study in imaging genomics* *Mar 2022*

15. Tech Talks 1.0, Student's Association of Artificial Intelligence and Data Science, G H Raisoni College of Engineering and Management, Pune, India (Virtual). *Computational biostatistics and biomedical imaging data analysis* *Feb 2022*

14. The Fifth Eastern Asia Chapter—International Society for Bayesian Analysis Conference: Satellite Meeting of the 2020 ISBA World Meeting to Celebrate James O. Berger's 70th Birthday (Virtual). *RADIOHEAD: radiogenomic analysis incorporating tumor heterogeneity in imaging through densities* *Nov 2021*

13. BU Biostatistics Student Association Seminar, BU. *RADIOHEAD: radiogenomic analysis incorporating tumor heterogeneity in imaging through densities* *Nov 2021*

12. Joint Statistical Meetings (Virtual). *Layered variable selection for multivariate Bayesian regression: a case study in imaging genomics* *Aug 2021*

11. Center for Computational Mathematics Seminar, Flatiron Institute, Simons Foundation,

New York (Virtual). *RADIOHEAD: radiogenomic analysis incorporating tumor heterogeneity in imaging through densities* *Jun 2021*

10. Statistical Methods in Imaging Conference (Virtual). *RADIOHEAD: radiogenomic analysis incorporating tumor heterogeneity in imaging through densities* *May 2021*
9. ENAR Spring Meeting (Virtual). *Layered variable selection for multivariate Bayesian regression: a case study in imaging genomics* *Mar 2021*
8. Precision Health Seminar (Pharmacy 217), U-M (Virtual). *Density-based classification using images: application to diabetic retinopathy* *Feb 2021*
7. U-M Precision Health Symposium (Virtual; Invited poster). *RADIOHEAD: radiogenomic analysis incorporating tumor heterogeneity in imaging via densities* *Aug 2020*
6. Joint Statistical Meetings (Virtual). *RADIOHEAD: radiogenomic analysis incorporating tumor heterogeneity in imaging through densities* *Aug 2020*
5. Michigan Institute for Data Science (MIDAS) COVID-19 Special Seminar Series, U-M (Group presentation). *Understanding COVID-19 dynamics via individual-level temporal and network modeling: lessons from India and China* *Jun 2020*
4. Precision Health Seminar (Pharmacy 217), U-M. *Integrative statistical modeling approaches for imaging genomics data* *Mar 2020*
3. Tools and Technology Seminar, U-M. *Integrative statistical modeling approaches for imaging genomics data* *Mar 2020*
2. ENAR Spring Meeting, Nashville, Tennessee (Virtual). *RADIOHEAD: radiogenomic analysis incorporating tumor heterogeneity in imaging through densities* *Mar 2020*
1. International Indian Statistical Association Annual Conference, Mumbai, India. *RADIOHEAD: radiogenomic analysis incorporating tumor heterogeneity in imaging through densities* *Dec 2019*

Invited Panels and Workshops

3. Advanced Research Institute (ARI) in Mental Health and Aging Spring Retreat, Winston-Salem, North Carolina (Invited participant). *Apr 2022*
2. Department of Biostatistics, BU (Panel). *Biostatistics career development panel: "Academic career path"* *Oct 2021*
1. NMIMS Sunandan Divatia School of Science (Virtual). *StatChat 2020: panel discussions with undergraduate and master's students on career and academic perspectives in statistics and data science* *Aug and Sep 2020*

Contributed Talks

12. Digital Cognitive Data Consortium, BU. *Predicting cognitive impairment using novel digital drawing features* *May 2026*
11. 65th ISI World Statistics Congress, The Hague, Netherlands. *Bayesian double generalized beta regression for ZIP code-level well-being assessment* *Oct 2025*
10. Symposium on Data Science and Statistics, Salt Lake City, Utah. *Modeling well-being using ZIP code spatial neighborhood structure in Bayesian beta regression* *Apr 2025*
9. BU Department of Biostatistics Research Blitz, BU. *Statistical and machine learning methods for complex data applications* *Mar 2023*
8. BU Department of Biostatistics Research Blitz, BU. *Statistical methods for biomedical imaging and digital data analysis* *Mar 2022*
7. ENAR Spring Meeting, Philadelphia, Pennsylvania. *Bayesian variable selection using spike-and-slab prior with application to high-dimensional EEG data* *Mar 2019*
6. Joint Statistical Meetings, Vancouver, Canada. *Bayesian variable selection using spike-and-slab prior with application to high-dimensional EEG data* *Jul 2018*
5. Symposium on Data Science and Statistics, Reston, Virginia. *Bayesian variable selection using spike-and-slab prior with application to high-dimensional EEG data* *May 2018*
4. Bayes Comp 2018, Barcelona, Spain (Poster). *Bayesian variable selection using spike-and-slab prior with application to high-dimensional EEG data* *Mar 2018*
3. International Indian Statistical Association Annual Conference, Hyderabad, India. *Bayesian variable selection using spike-and-slab prior for EEG analysis* *Dec 2017*
2. 34th Quality and Productivity Research Conference, UConn. *Bayesian variable selection using spike-and-slab prior with application to high-dimensional EEG data* *Jun 2017*
1. 31st New England Statistics Symposium, UConn. *Modeling of large insurance claims and occurrence data* (joint with Aritra Halder) *Apr 2017*

SERVICE & LEADERSHIP

Professional Service

Editorial

- *Editorial Board Member*, Scientific Reports *2026–present*
- *Associate Editor*, Sankhyā B: The Indian Journal of Statistics (Official Journal of Indian Statistical Institute) *2023–present*

Professional Societies

- *Assistant Executive Administrator*, International Indian Statistical Association (IISA) *2026–present*
- *Vice-President*, Conferences Committee, New England Statistics Society (NESS) *2024–present*
- *Founding Co-Organizer*, IISA Statistics and Data Science Innovations Webinar Series *2022*
- *Member*, Membership & Outreach Committee, IISA *2020–2022*

Conference Organization

- *Member*, Conference Advisory Committee, New England Statistics Symposium *2025*
- *Member*, Program Committee, New England Statistics Symposium *2024*
- *Member*, Scientific Organizing Committee, Red Rock Data Science Student Conference *2024*
- *Co-Chair*, Organizing Committee, New England Statistics Symposium (host: Boston University) *2022–2023*
- *Co-Vice President for Scientific Program*, New England Statistics Symposium *2022–2023*
- *Member*, Organizing Committee, *Pushing the Boundary of Data Science through Statistical Modeling and Inference* (Conference in honor of Dipak K. Dey) *2022–2023*
- *Member*, Student Research Awards Committee, New England Statistics Symposium *2022*
- *Vice-Chair* of Student Committee, New England Statistics Symposium *2017*
- *Organizer (Invited Sessions)*: Joint Statistical Meetings (JSM) 2020; ENAR Spring Meeting 2021; JSM 2021; NESS 2022; JSM 2022; IISA Annual Conference 2022; JSM 2023; NESS 2024; International Biometric Conference (IBC) 2026

Grant, Award, and Abstract Review

- *Reviewer*, Student Paper Competition, American Statistical Association Section on Bayesian Statistical Science (SBSS) *2024–present*
- *Reviewer*, American Public Health Association Annual Meeting *2024–present*
- *Reviewer*, Framingham Heart Study Brain Aging Program Pilot Awards *2024*
- *Reviewer*, Grant Review and Mock Study Section, Advanced Research Institute *2022, 2024*

Journal Review

- *Ad-hoc Reviewer (Statistics/Biostatistics Journals)*: Annals of Applied Statistics; Bayesian Analysis; Biometrics; Biostatistics; Communications in Statistics – Theory and Methods; Harvard Data Science Review; Journal of Applied Statistics; The New England Journal of Statistics in Data Science; Sankhyā B; Spatial Statistics; STAT; Statistics and Data Science in Imaging

- *Ad-hoc Reviewer (Interdisciplinary Journals):* Advances in Biomarker Sciences and Technology; Bioinformatics; Clinical Cancer Informatics; Epidemiology; Frontiers in Digital Health; Journal of the American Heart Association; Journal of the American Medical Informatics Association; Scandinavian Actuarial Journal

University Service

Boston University: Department of Biostatistics

- *Reviewer*, BU Biostatistics Student Paper Competition *2025–present*
- *Member*, Biostatistics Qualifying Exam Committee, BU *2024–present*
- *Co-Organizer*, Biostatistics Department Seminar Series, BU *2021–present*

Boston University: School of Public Health

- *Member*, Environmental Health Doctoral Exam Committee for Beverly Ge at BU *2026*
- *Member*, Environmental Health Faculty Search Committee, BU *2022–2023*

Boston University: University-wide

- *Member*, Graduate Student Fellow Awards Committee, Rafik B. Hariri Institute for Computing and Computational Science & Engineering *2023*

University of Connecticut

- *Advisor / President / Treasurer*, Tarang (South Asian cultural organization), UConn *2015–2018*
- *Vice-President*, Statistics Graduate Student Committee, UConn *2016–2017*
- *Co-President*, Statistics Graduate Student Committee, UConn *2015–2016*
- *Senator*, UConn Graduate Student Senate *2015–2016*
- *Student Representative*, Senate Faculty Standards Committee, UConn *2015–2016*

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